Chapter 5, Section 2

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1. If X is a birationally ruled surface, show that the curve C, such that X is birationally equivalent to $C \times \mathbb{P}^1$, is unique up to isomorphism.

Proof. Let C and C' be two nonsingular curves such that X is birational to $C \times \mathbb{P}^1$ and $C' \times \mathbb{P}^1$, and suppose $\varphi : C \times \mathbb{P}^1 \to C' \times \mathbb{P}^1$ is a birational map with inverse ψ. The case is trivial if either C or C' has genus 0, so assume otherwise. For any fiber f of $C \times \mathbb{P}^1$, consider the composition map $p_1 \circ \varphi : f \to C'$. Since both f and C' are complete curves, $p_1 \circ \varphi$ extends to a morphism, and if C' has genus > 0, then it must be constant. Therefore, $p_1 \circ \varphi$ factors through $p_1 : C \times \mathbb{P}^1 \to C$, which extends to a morphism $C \to C'$. Doing the same thing for ψ gives an isomorphism of C with C'. □

2. Let X be the ruled surface $\mathbb{P}(\mathcal{E})$ over a curve C. Show that $\mathcal{E}$ is decomposable if and only if there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$.

Proof. If $\mathcal{E}$ is decomposable, then $\mathcal{E} = \mathcal{O} \oplus \mathcal{L}$, where $\mathcal{L} = \mathcal{O}_C(\mathfrak{e})$ for some divisor $\mathfrak{e} \in \operatorname{Div} C$. By (2.8.1), (2.11.1), and (2.11.2), the two natural projections $\mathcal{E} \to \mathcal{L}$ and $\mathcal{E} \to \mathcal{O}_C$ correspond to sections $C' \sim C_0$ and $C'' \sim C_0 - \mathfrak{e} f$, respectively, such that $C'.C'' = 0$. Hence, $C' \cap C'' = \emptyset$. Conversely, suppose there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$. The corresponding surjective morphisms $\mathcal{E} \to \mathcal{L}'$ and $\mathcal{E} \to \mathcal{L}''$ induce an $\mathcal{O}_C$-module homomorphism $\mathcal{E} \to \mathcal{L}' \oplus \mathcal{L}''$. For every $P \in C$, the morphism of stalks $\mathcal{E}_P \to \mathcal{L}'_P \oplus \mathcal{L}''_P$ is an isomorphism because C' and C'' are disjoint, so $\mathcal{L}'_P \oplus \mathcal{L}''_P$ is generated by $\mathcal{E}_P$. Hence, $\mathcal{E} \to \mathcal{L}' \oplus \mathcal{L}''$ is an isomorphism. □

3. (a) If $\mathcal{E}$ is a locally free sheaf of rank r on a nonsingular curve C, then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that $\mathcal{E}_i / \mathcal{E}_{i-1}$ is an invertible sheaf for each $i = 1, \ldots, r$. We say that $\mathcal{E}$ is a *successive extension* of invertible sheaves.

(b) Show that this is false for varieties of dimension ≥ 2. In particular, the sheaf of differentials $\Omega_{\mathbb{P}^2}$ on $\mathbb{P}^2$ is not an extension of invertible sheaves.

Proof.

(a) We proceed by induction on r. The base case $r = 1$ is trivial, so assume the statement for any $r > 1$. We can twist $\mathcal{E}$ by a very ample sheaf $\mathcal{L}$ so that $\mathcal{E}' = \mathcal{E} \otimes \mathcal{E}$ is generated by global sections. If we prove the statement for $\mathcal{E}'$ so we have a filtration $(\mathcal{E}'_i)$, then tensoring by $\mathcal{L}^{-1}$ gives a filtration $(\mathcal{E}_i = \mathcal{E}'_i \otimes \mathcal{L}^{-1})$ of $\mathcal{E}$ such that

$$\mathcal{E}_i / \mathcal{E}_{i-1} \cong (\mathcal{E}'_i / \mathcal{E}'_{i-1}) \otimes \mathcal{L}^{-1},$$

which is an invertible sheaf for each $i = 1, \ldots, r$. Hence, we can assume $\mathcal{E}$ is generated by global sections. The hypothesis of (II, Ex. 8.2) is satisfied for $\mathcal{E}$, so there exists a short exact

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0,$$

where $\mathcal{F}$ is also globally generated locally free of rank $r - 1$. By the inductive hypothesis, this means we can find a filtration $(\mathcal{F}_i)$ of $\mathcal{F}$ such that each successive quotient is an invertible sheaf. Consider the preimage $(\mathcal{E}_j)$ of $(\mathcal{F}_i)$, where we relabel $j = i + 1$. Each $\mathcal{E}_j$ is also locally free, $\mathcal{E}_1 = \operatorname{im}(\mathcal{O}_C)$ by exactness, and there is a canonical isomorphism $\mathcal{E}_j / \mathcal{E}_{j-1} \cong \mathcal{F}_{j-1} / \mathcal{F}_{j-2}$. Extending $(\mathcal{E}_j)$ with $\mathcal{E}_0 = 0$ gives a successive extension of sheaves.

(b) Suppose we have an exact sequence

$$0 \longrightarrow \mathcal{L} \longrightarrow \Omega \longrightarrow \mathcal{M} \longrightarrow 0,$$

where $\mathcal{L}$ and $\mathcal{M}$ are invertible sheaves on $\mathbb{P}^2$. Then $\mathrm{H}^1(\mathcal{L}), \mathrm{H}^1(\mathcal{M}) = 0$, so by the long exact sequence in cohomology, we have $\mathrm{H}^1(\Omega) = 0$. But by (III, Ex. 7.2), $\mathrm{H}^1(\Omega) = k$, a contradiction.

□

4. Let C be a curve of genus g, and let X be the ruled surface $C \times \mathbb{P}^1$. We consider the question, for what integers $s \in \mathbb{Z}$ does there exist a section D of X with $D^2 = s$? First show that s is always an even integer, say $s = 2r$.

(a) Show that $r = 0$ and $r \geq g + 1$ are always possible.

(b) If $g = 3$, show that $r = 1$ is not possible, and just one of the two values $r = 2, 3$ is possible, depending on whether C is hyperelliptic or not.

Proof. Fix a section $C_0 = C \times P_0 \subset X$, where P_0 is a point of $\mathbb{P}^1$. The ruled surface $X = C \times \mathbb{P}^1$ corresponds to the projective bundle of $\mathcal{E} = \mathcal{O}_C \oplus \mathcal{O}_C$. The invariant of X is $e = 0$, which means $C_0^2 = -e = 0$. By (2.9), if D is any section of X corresponding to a surjection $\mathcal{E} \to \mathcal{O}_C(\mathfrak{d})$ for some divisor $\mathfrak{d}$ of C with $r = \deg \mathfrak{d}$, then $r = C_0.D$ and $D \sim C_0 + \mathfrak{d}f$. Hence, $s = 2r$.

(a) There is a one-to-one correspondence between sections D of X and finite separable morphisms $f : C \to \mathbb{P}^1$. In particular, a section D is the graph of a morphism $f : C \to \mathbb{P}^1$, and its self-intersection is $D^2 = 2{\cdot}\deg f$. By (IV, Ex. 1.6), $g + 1$ is an upperbound on the minimum of $\deg f$. Hence, $r = 0$ and $r \geq g + 1$ is always possible.

(b) If $g > 0$, then there is no morphism $C \to \mathbb{P}^1$ of degree 1. If C is a hyperelliptic curve, then it has a g^1_2, so $r = 2, 3$ is possible. Otherwise, nonhyperelliptic curves of genus 3 always have a g^1_3 (IV, 5.5.2), so $r = 3$ is always possible.

□

5. *Values of e.* Let C be a curve of genus $g \geq 1$.

(a) Show that for each $0 \leq e \leq 2g - 2$ there is a ruled suface X over C with invariant e, corresponding to an indecomposable $\mathcal{E}$.

(b) Let $e < 0$, let D be any divisor of degree $d = -e$, and let $\xi \in \mathrm{H}^1(X, \mathcal{O}(-D))$ be a nonzero element defining an extension

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(D) \longrightarrow 0.$$

Let $H \subseteq |D + K|$ be the sublinear system of codimension 1 defined by $\ker \xi$, where ξ is considered as a linear functional on $\mathrm{H}^0(\mathcal{O}(D + K))$. For any effective divisor E of degree $d - 1$, let $L_E \subseteq |D + K|$ be the sublinear system $|D + K - E| + E$. Show that $\mathcal{E}$ is normalized if and only if for each E as above, $L_E \not\subseteq H$.

(c) Now show that if $-g \leq e < 0$, there exists a ruled surface X over C with invariant e.

(d) For $g = 2$, show that $e \geq -2$ is also necessary for the existence of X.

Proof.

(a) Let $\mathfrak{e}$ be a divisor of degree $-e$ on C, and let $\mathfrak{k}$ is a canonical divisor of C. We need to first construct an indecomposable extension $\mathcal{E}$ of $\mathcal{O}(\mathfrak{e})$ by $\mathcal{O}$. By (III, Ex. 6.1), an extension of vector bundles

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(\mathfrak{e}) \longrightarrow 0$$

corresponds to an element in $\mathrm{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O})$. In this case, $\mathrm{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O}) \cong \mathrm{H}^1(\mathcal{O}(-\mathfrak{e}))$ (III, 6.3) and (III, 6.7). This is dual to $\mathrm{H}^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e}))$, so it is enough to find $\mathfrak{e}$ such that $\mathrm{H}^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e})) \neq 0$. The image of the canonical morphism $C \to |\mathfrak{k}| \cong \mathbb{P}^{g-1}$ is either a rational curve of degree $g - 1$, or an embedding of degree $2g - 2$, depending on whether or not C is hyperelliptic. If $-\mathfrak{e}$ is effective, then $\mathrm{H}^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e}))$ is the space

of all hyperplane sections in $\mathbb{P}^{g-1}$ with support containing $\mathfrak{e}$. In any case, a hyperplane section of C in $\mathbb{P}^{g-1}$ consists of $2g-2$ points counted with multiplicity. Hence, for each $0 \le e \le 2g-2$, there is a divisor $\mathfrak{e}$ of degree $-e$ such that $\mathrm{H}^0(\mathcal{O}(\mathfrak{k}+\mathfrak{e})) \neq 0$, so there exists indecomposable extensions of $\mathcal{O}(\mathfrak{e})$ by $\mathcal{O}$.
Now, let $\mathcal{E}$ be an extension of $\mathcal{O}(\mathfrak{e})$ defined by a nonzero element in $\mathrm{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O})$. It remains to show $\mathcal{E}$ is normalized and indecomposable. Clearly $\mathrm{H}^0(\mathcal{E}) \neq 0$ by construction. If $\mathcal{M}$ is any invertible sheaf, then we have an exact sequence

$$0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{E} \otimes \mathcal{M} \longrightarrow \mathcal{O}(\mathfrak{e}) \otimes \mathcal{M} \longrightarrow 0.$$

If $\deg \mathcal{M} < 0$, then we have $\mathrm{H}^0(\mathcal{M}) = 0$ and $\mathrm{H}^0(\mathcal{O}(\mathfrak{e}) \otimes \mathcal{M}) = 0$, and therefore also $\mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) = 0$. Furthermore, if $\mathcal{E}$ is decomposable, then it would be $\mathcal{O} \oplus \mathcal{L}$ for some $\mathcal{L}$, by (2.12). But $\bigwedge^2 \mathcal{E} \cong \mathcal{O}_C(\mathfrak{e})$, so $\mathcal{L} \cong \mathcal{O}(\mathfrak{e})$, so in fact this extension would have to split, which it doesn't. Hence, for each $0 \le e \le 2g-2$, there is a ruled surface X over C with invariant e, corresponding to an indecomposable $\mathcal{E}$.

(b) Let $\xi \in \mathrm{H}^1(\mathcal{O}(-D))$ be the (non-zero) element defining the extension $\mathcal{E}$. If E is any effective divisor of degree $d-1$ on C, then we have a commutative diagram

$$\begin{array}{ccc} \mathrm{H}^0(\mathcal{O}) & \xrightarrow{\delta} & \mathrm{H}^1(\mathcal{O}(-D))) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} \\ \mathrm{H}^0(\mathcal{O}(E)) & \xrightarrow{\gamma} & \mathrm{H}^1(\mathcal{O}(E-D)) \end{array}$$

where $\delta(1) = \xi$, $\alpha(1) = t$, a non-zero section defining E, and β is induced from the map $\mathcal{O} \to \mathcal{O}(E)$ corresponding to t. On the other hand, β is dual to the map

$$\beta^\vee : \mathrm{H}^0(\mathcal{O}(D+K-E)) \to \mathrm{H}^0(\mathcal{O}(D+K))$$

also induced by t. The subspace of $\mathrm{H}^0(\mathcal{O}(D+K))$ corresponding to L_E is precisely the image of $\beta^\vee$, so $\beta(\xi) = 0$ if and only if $L_E \subseteq H$.
If $\mathcal{E}$ is normalized, then γ is injective for any effective divisor E of degree $d-1$. By the commutative diagram, we must have $\beta(\xi) \neq 0$. Hence, $L_E \not\subseteq H$.
Conversely, if $\mathcal{E}$ is not normalized, then $\mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) \neq 0$ for some invertible sheaf $\mathcal{M}$ with $\deg \mathcal{M} < 0$, and let $m = -\deg \mathcal{M}$. We have the exact sequence

$$0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{E} \otimes \mathcal{M} \longrightarrow \mathcal{O}(D) \otimes \mathcal{M} \longrightarrow 0 \ .$$

If $m \ge d$, then we have $\mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) = 0$ since $\deg \mathcal{O}(D) \otimes \mathcal{M} < 0$, and if $m = d$, then either $\mathrm{H}^0(\mathcal{O}(D) \otimes \mathcal{M}) = 0$, or $\mathcal{M} = \mathcal{O}(-D)$, whence the coboundary map δ is injective, hence $\mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) = 0$. So we can assume $0 < m < d$. The long exact sequence of cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) \longrightarrow \mathrm{H}^0(\mathcal{O}(D) \otimes \mathcal{M}) \longrightarrow \mathrm{H}^1(\mathcal{M}) \ ,$$

where $\mathrm{H}^0(\mathcal{M}) = 0$. The middle term is non-zero, so if $\mathcal{M} \cong \mathcal{O}(-M)$ with $m = \deg M$, then $D - M \sim E$ for some effective divisor E of degree $d-m$. For any effective divisor $\tilde{E}$ of degree $d-1$ with $\tilde{E} - E \ge 0$, we have a commutative diagram,

$$\begin{array}{ccc} \mathrm{H}^0(\mathcal{O})) & \xrightarrow{\delta} & \mathrm{H}^1(\mathcal{O}(-D)) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} \\ \mathrm{H}^0(\mathcal{O}(E)) & \xrightarrow{\gamma} & \mathrm{H}^1(\mathcal{O}(E-D)) \\ \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle \psi} \\ \mathrm{H}^0(\mathcal{O}(\tilde{E})) & \xrightarrow{\eta} & \mathrm{H}^1(\mathcal{O}(\tilde{E}-D)) \end{array} \qquad \tilde{\alpha}, \ \tilde{\beta},$$

where $\delta(1) = \xi$, α, β is induced by a section $t \in \mathrm{H}^0(\mathcal{O}(E))$ defining E, and ϕ, ψ is induced by a section s defining $\tilde{E} - E$.
Now, if $\beta(\xi) = 0$, then we are done. Otherwise, dualizing the column on the right of the commutative diagram gives

$$\begin{array}{ccc} \mathrm{H}^0(\mathcal{O}(D+K-\tilde{E})) & & \\ \downarrow{\scriptstyle \psi^\vee} & \searrow^{\tilde{\beta}^\vee} & \\ \mathrm{H}^0(\mathcal{O}(D+K-E)) & \xrightarrow{\beta^\vee} & \mathrm{H}^0(\mathcal{O}(D+K)) \end{array} \ .$$

Thus, for every $\tilde{E}$, we have $L_{\tilde{E}} \subseteq L_E$. By the assumption $\beta(\xi) \neq 0$, we have $L_E \not\subseteq H$, so $\mathrm{im}(\beta^\vee) \cap \ker(\xi)$ is a nonempty proper subspace of $\mathrm{im}(\beta^\vee)$. By the Riemann-Roch theorem, we have

$$\dim \mathrm{H}^0(\mathcal{O}(D+K-\tilde{E})) = g, \quad \dim \mathrm{H}^0(\mathcal{O}(D+K-E)) = g+m-1.$$

The space of possible $\tilde{E}$ has dimension $m-1$, so the image of $\psi^\vee$ as $\tilde{E}$ varies is dense inside $\mathrm{H}^0(\mathcal{O}(D+K-E))$. Consequently, we can find at least one $\tilde{E}$ such that $\mathrm{im}(\tilde{\beta}^\vee) \not\subseteq \ker(\xi)$, which is what we wanted to show.

(c) Let $\mathbb{P}^N = |K+D|$ with $N = g+d-2$. For any effective divisor E of degree $d-1$ on C, let us consider the set

$$B_E = \{\text{hyperplanes } H \subset \mathbb{P}^N \mid L_E \subseteq H\} \subset |H|,$$

where $|H|$ is the complete linear system of hyperplanes in $\mathbb{P}^N$, i.e., $|H| = |K+D|$. A hyperplane in $\mathbb{P}^N$ is determined by a nonzero global section $f \in V = \mathrm{H}^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$. Viewing L_E as a linear subspace of $\mathbb{P}^N$, let $W_E = \mathrm{H}^0(L_E, \mathcal{O}_{L_E}(1))$. The space of effective divisors of degree $d-1$ is parameterized by the closed points of the symmetric product $C^{(d-1)}$, which has dimension $d-1$. Consider the restriction map

$$\varphi_E : V \to W_E.$$

It is clearly surjective, and the hyperplanes $H \in B_E$ correspond exactly to those $f \in \ker \varphi_E$. We have $\dim V = g+d-1$ and $\dim W_E = g$, so $\dim B_E = \dim(\ker \varphi_E) - 1 = d-2$.

Now, consider the subset $B \subset C^{(d-1)} \times |H|$ consisting of all pairs $\langle E, H\rangle$ such that $E \in C^{(d-1)}$ is a closed point and $H \in B_E$. Clearly B is the set of closed points of a $C^{(d-1)} \times |H|$, which we denote also by B, and which we give a reduced induced scheme structure. The first projection $p_1 : B \to C^{(d-1)}$ is surjective, with fibre a projective space of dimension $d-2$. So B has dimension $(d-1)+(d-2) = 2d-3$, which is strictly smaller than $\dim |H| = g+d-2$. If $H \in |H| - p_2(B)$, then $H \not\supseteq L_E$ for every $E \in C^{(d-1)}$, so the extension of $\mathcal{O}(D)$ by $\mathcal{O}$ given by the element in $\mathrm{H}^0(\mathcal{O}(K+D)) \cong \mathrm{H}^1(\mathcal{O}(-D))$ corresponding to H satisfies the requirement of (b). Hence, if $-g \leq e < 0$, then there exists a ruled surface X over C with invariant e.

(d) Let C be a curve of genus $g = 2$, let $\mathcal{E}$ be a normalized locally free sheaf of rank 2 on C with invariant e, let $d = -e$, and let $X = \mathbb{P}(\mathcal{E})$ be the ruled surface associated to $\mathcal{E}$. As in the proof of (2.15), we want to show $e = -3, -4$ does not occur.

If it did, we would have a normalized bundle $\mathcal{E}$ with an exact sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(D) \longrightarrow 0 \ ,$$

where D is an effective divisor of degree $d = -e$. Let $\xi \in \mathrm{H}^1(\mathcal{O}(-D))$ be the element defining the extension $\mathcal{E}$. If E is any effective divisor, then applying $\mathrm{Hom}(\mathcal{O}(D-E), -)$ and taking the long exact sequence in cohomology gives

$$\mathrm{Hom}(\mathcal{O}(D-E), \mathcal{E}) \longrightarrow \mathrm{Hom}(\mathcal{O}(D-E), \mathcal{O}(D)) \xrightarrow{\delta_E} \mathrm{Ext}^1(\mathcal{O}(D-E), \mathcal{O}) \ ,$$

so by exactness, the inclusion $i : \mathcal{O}(D-E) \to \mathcal{O}(D)$ factors through $\mathcal{E} \to \mathcal{O}(D)$ if and only if $\delta_E(i) = 0$. By naturality, we have a commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}(\mathcal{O}(D), \mathcal{O}(D)) & \xrightarrow{\delta} & \mathrm{Ext}^1(\mathcal{O}(D), \mathcal{O}) \\ \downarrow{\scriptstyle i^*} & & \downarrow{\scriptstyle i^*} \\ \mathrm{Hom}(\mathcal{O}(D-E), \mathcal{O}(D)) & \xrightarrow{\delta_E} & \mathrm{Ext}^1(\mathcal{O}(D-E), \mathcal{O})) \end{array} \ ,$$

where $\delta(1) = \xi$ and $i^*(1) = i$, so $\delta_E(i) = 0$ if and only if $i^*(\xi) = 0$. Now, if there exists E of degree $< d$ such that $i : \mathcal{O}(D-E) \to \mathcal{E}$ factors through $\mathcal{E} \to \mathcal{O}(D)$, then $\mathcal{O}(D-E) \to \mathcal{E}$ must also be injective, which is a contradiction of $\mathcal{E}$ being normalized, so it is enough to show there exists E such that $i^*(\xi) = 0$. The discussion above has the following geometric interpretation. By Serre duality, there is a natural identification

$$|D+K| \cong \mathbb{P}(\mathrm{Ext}^1(\mathcal{O}(D), \mathcal{O})) \cong \mathbb{P}(\mathrm{H}^0(\mathcal{O}(D+K))^\vee),$$

so we can view the linear span of ξ in $\mathrm{Ext}^1(\mathcal{O}(D), \mathcal{O})$ as a point in $|D+K|$, which we will also denote by ξ. It follows that $i^*(\xi) = 0$ if and only if ξ is contained in every hyperplane in $|D+K|$ which also contains E.

$\boxed{d=3}$ In either case, $D+K$ is a very ample divisor on C (IV, Ex. 3.1), so consider the embedding of C in $|D+K| \cong \mathbb{P}^3$. The secant variety of C fills up the entire space (IV, §3), so there is a secant line $L \subseteq |D+K|$ which contains ξ. A line is uniquely determined by two points, so we can take E to be any effective divisor of degree 2 with support contained in $C \cap L$. Hence, any hyperplane $H \subseteq |D+K|$ containing E also contains L, so ξ must be a point in H as well.

$\boxed{d=4}$ Again, consider C in its embedding in $|D+K| \cong \mathbb{P}^4$. A plane is uniquely determined by three points, and the union of all planes going through at least three points of C fills up all of $|D+K|$. This means we can find a plane P in $|D+K|$ which contains ξ and at least three points of C. So we can take E to be any effective divisor of degree 3 with support contained in $C \cap P$. Hence, any hyperplane $H \subseteq |D+K|$ containing E also contains P, so ξ must be in H as well.

□

6. Show that every locally free sheaf of finite rank on $\mathbb{P}^1$ is isomorphic to a direct sum of invertible sheaves.

Proof. We call any locally free sheaf $\mathcal{E}$ on a curve C *normalized* if $\mathrm{H}^0(C,\mathcal{E}) \neq 0$ and $\mathrm{H}^0(C,\mathcal{E}\otimes\mathcal{L}) = 0$ for all invertible sheaves $\mathcal{L}$ on C with $\deg\mathcal{L} < 0$. With this definition, the statement of (2.8) can be made sense for $\mathcal{E}$ of arbitrary rank:

Lemma 0.1. *For every locally free sheaf $\mathcal{E}$ of finite rank on a curve C, there exists an invertible sheaf $\mathcal{L}$ on C such that $\mathcal{E}\otimes\mathcal{L}$ is normalized.*

Going back to the original statement, we can assume $\mathcal{E}$ is normalized. We proceed by induction on the rank r of $\mathcal{E}$. The base case $r=1$ is trivial, so assume the statement holds for any $< r$. Pick a subinvertible sheaf $0 \to \mathcal{O} \to \mathcal{E}$ of maximal degree, and consider the short exact sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0\ .$$

The quotient sheaf $\mathcal{F} = \mathcal{E}/\mathcal{O}$ is localy free of rank $r-1$ because of the choice of $0 \to \mathcal{O} \to \mathcal{E}$ (2.8), so by the inductive hypothesis, $\mathcal{F}$ is decomposable, say $\mathcal{F} = \bigoplus \mathcal{O}(b_i)$ for some $b_i \in \mathbb{Z}$. Tensoring by $\mathcal{O}(-1)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathcal{O}(-1)) \longrightarrow \mathrm{H}^0(\mathcal{E}(-1)) \longrightarrow \mathrm{H}^0(\textstyle\bigoplus_i \mathcal{O}(b_i-1)) \longrightarrow 0\ .$$

Since $\mathcal{E}$ is normalized, the middle vector space is zero, so by exactness we have $b_i \leq 0$. On the other hand, the space of extensions of $\mathcal{F}$ by $\mathcal{O}$ can be computed by Serre duality, which gives

$$\mathrm{Ext}^1(\mathcal{F},\mathcal{O}) \cong \mathrm{H}^0(\textstyle\bigoplus \mathcal{O}(b_i-1)) = 0,$$

so any extension of $\mathcal{F}$ by $\mathcal{O}$ is trivial. Hence, the short exact sequence splits, and there is an isomorphism $\mathcal{E} \cong \mathcal{O}\oplus\mathcal{F}$. □

7. On the elliptic ruled surface X of (2.11.6), show that the sections C_0 with $C_0^2 = 1$ form a one-dimensional algebraic family, parametrized by the points of the base curve C, and that no two are linearly equivalent.

Proof. Combine (III, Ex. 11.5), (2.9), (2.16) and the fact that any divisor of positive degree on an elliptic curve is linearly equivalent to an effective divisor. The correspondence is given by for each closed point $Q \in C$, there is a section linearly inequivalent to $C_0 + (Q-P)f$, i.e., there is a surjection $\mathcal{E} \to \mathcal{O}_C(Q-P)$ of $\mathcal{O}_C$-modules. By (2.1), $\pi_*(\mathcal{O}_X(C_0 + (Q-P)f))$ is isomorphic to $\mathcal{E}\otimes\mathcal{O}_C(Q-P)$, and by (2.16) these locally free sheaves of rank 2 are in one-to-one corresponding with the closed points of C. □

8. A locally free sheaf $\mathcal{E}$ on a curve C is said to be *stable* if for every quotient locally free sheaf $\mathcal{E} \to \mathcal{F} \to 0$, $\mathcal{F} \neq \mathcal{E}$, $\mathcal{F} \neq 0$, we have

$$(\deg\mathcal{F})/\operatorname{rank}\mathcal{F} > (\deg\mathcal{E})/\operatorname{rank}\mathcal{E}$$

Replacing $>$ by $\geq$ defines *semistable.*

(a) A decomposable $\mathcal{E}$ is never stable.

(b) If $\mathcal{E}$ has rank 2 and is normalized, then $\mathcal{E}$ is stable (respectively, semistable) if and only if $\deg\mathcal{E} > 0$ (respectively, ≥ 0).

(c) Show that the indecomposable locally free sheaves $\mathcal{E}$ of rank 2 that are not semistable are classified, up to isomorphism, by giving (1) an integer $0 < e \leq 2g-2$, (2) an element $\mathcal{L} \in \operatorname{Pic} C$ of degree $-e$, and (3) a nonzero $\xi \in \mathrm{H}^1(C, \mathcal{L}^{-1})$, determined up to a nonzero scalar multiple.

Proof.

(a) Denote $\mu(\mathcal{E}) = (\deg \mathcal{E})/\operatorname{rank} \mathcal{E}$ the slope of $\mathcal{E}$, and let $\mathcal{F} \subseteq \mathcal{E}$ consist of the summands of $\mathcal{E}$ that has degree $\leq \mu(\mathcal{E})$, which there must be at least one. There is a surjective morphism of locally free sheaves $\mathcal{E} \to \mathcal{F} \to 0$, and we have $\mu(\mathcal{F}) \leq \mu(\mathcal{E})$. Hence, $\mathcal{E}$ is not stable.

(b) If $\mathcal{E}$ is normal with invariant $e = -\deg \mathcal{E}$, then we have a quotient locally sheaf $\mathcal{E} \to \mathcal{L} \to 0$, where $\mathcal{L}$ is an invertible sheaf of degree $-e$. If $\mathcal{E}$ is stable (respectively, semistable), then

$$-e = \mu(\mathcal{L}) > \mu(\mathcal{E}) = -e/2,$$

(respectively, $\mu(\mathcal{L}) \geq \mu(\mathcal{E})$), which implies $\deg \mathcal{E} = -e > 0$ (respectively, ≥ 0). Conversely, suppose $\deg \mathcal{E} > 0$ (respectively, ≥ 0), and let $\mathcal{E} \to \mathcal{F} \to 0$ be a quotient locally free sheaf. If $\mathcal{F}$ has rank 2, then $\mathcal{E} \to \mathcal{F} \to 0$ is an isomorphism (II, Ex. 7.1). Otherwise, $\mathcal{F}$ is an invertible sheaf, and because of the normalization of $\mathcal{E}$, we must have $\deg \mathcal{F} \geq \deg \mathcal{E}$. Hence, $\mu(\mathcal{F}) = \deg \mathcal{F} \geq \deg \mathcal{E} > \mu(\mathcal{E})$ (respectively, $\geq \mu(\mathcal{E})$) if $\deg \mathcal{E} > 0$ (respectively, $\deg \mathcal{E} \geq 0$).

(c) There is a small typo in the problem statement. The classification should concern *normalized* indecomposable locally free sheaves of rank 2 that are not semistable. Combine (III, Ex. 6.1), (III, Ex. 12.5), (b), and the fact that the extension $0 \to \mathcal{O} \to \mathcal{E} \to \mathcal{L} \to 0$ defined by any $\xi \in \mathrm{H}^1(C, \mathcal{L}^{-1})$ is normalized if $e > 0$.

□

9. Let Y be a nonsingular curve on a quadric cone X_0 in $\mathbb{P}^3$. Show that either Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, in which case $\deg Y = 2a$, $g(Y) = (a-1)^2$, or $\deg Y$ is odd, say $2a+1$, and $g(Y) = a^2 - a$.

Proof. A general hyperplane section of X_0 in $\mathbb{P}^3$ is a conic curve C, and X_0 is a cone over C in $\mathbb{P}^3$ with vertex P_0. If we blow up the point P_0 as in (2.11.4), then we obtain the rational ruled surface $\mathbb{F}$ with invariant $e = 2$, such that if $\pi : \mathbb{F} \to X_0$ is the projection map, then $\pi^{-1}(P_0)$ is a section C_0 with $C_0^2 = -2$. In other words, $\mathbb{F}$ is isomorphic to the projective bundle of $\mathcal{E} = \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ over $\mathbb{P}^1$. The hyperplane sections of $\pi : \mathbb{F} \to X_0 \hookrightarrow \mathbb{P}^3$ correspond to the linear system $|\mathcal{O}_{\mathbb{F}}(1)| = |C_0 + 2f|$, and by (2.11), $K = -2C_0 - 4f$ is a canonical divisor of $\mathbb{F}$.

If Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, then $Y \sim a(C_0 + 2f)$, so $\deg Y = Y.(C_0 + 2f) = 2a$, and the adjunction formula gives

$$g = \frac{1}{2} Y.(Y + K) + 1 = (a-1)^2.$$

Conversely, if Y has even degree, say $\deg Y = 2a$, then $Y \sim a(C_0 + 2f) - bC_0$ with $0 \leq b < a$ (1.3) (2.20). Let $\mathcal{I} = \mathcal{O}_{\mathbb{F}}(-Y)$ be the ideal sheaf of Y as a closed subscheme of $\mathbb{F}$. We can tensor by $\mathcal{O}_{\mathbb{F}}(a)$, take global sections, and apply (2.4), which gives

$$\mathrm{H}^0(\mathbb{F}, \mathcal{I}(a)) \cong \mathrm{H}^0(\mathbb{P}^1, S^b(\mathcal{E})),$$

where we have

$$S^b(\mathcal{E}) \cong \bigoplus_{i=0}^{b} \mathcal{O}_{\mathbb{P}^1}(-2i).$$

It follows that there exists a section in $|\mathcal{O}_{\mathbb{F}}(a)|$ containing Y, and since the restriction map

$$\mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(a)) \to \mathrm{H}^0(\mathbb{F}, \mathcal{O}_{\mathbb{F}}(a)) \to 0$$

is surjective, we can find a homogenous polynomial of degree a in $\mathbb{P}^3$ containing Y. So by reasons of degree, Y is the complete intersection of X_0 with a surface of degree a.

If Y has odd degree, say $\deg Y = 2a+1$, then Y is contained in a surface of degree $a+1$. If Z is any such surface in $\mathbb{P}^3$, by (I, 7.7) $X_0 \cap Z$ has two irreducible components each of multiplicity one, consisting of Y

and a line L in $\mathbb{P}^3$. The latter must be a member of the one-dimensional family of lines in X_0 which go through the vertex P_0, and the inverse image $\pi^*L = L \times_{X_0} X$ of any of these lines is linearly eqiuvalent to $C_0 + f$, which means

$$Y \sim (a+1)(C_0 + 2f) - (C_0 + f) = aC_0 + (2a+1)f.$$

To conclude, we can compute the genus of Y using the adjunction formula of (1.5)

$$g = \frac{1}{2}Y.(Y + K) + 1 = a^2 - a.$$

□

10. For any $n > e \geq 0$, let X be the rational scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$ given by (2.19). If $n \geq 2e - 2$, show that X contains a nonsingular curve Y of genus $g = d + 2$ which is a canonical curve in this embedding. Conclude that for every $g \geq 4$, there exists a nonhyperelliptic curve which has a g^1_3.

Proof. Let D be the divisor $C_0 + nf$ which gives the emebdding of X as a rational scrolled of degree d in $\mathbb{P}^{d+1}$. If Y is a nonsingular curve in X of genus $g = d + 2$ and of type (a, b) satisfying (2.18), then $Y.D$ is the degree of Y under this embedding, and the genus of Y can computed using the adjunction formula (1.5). For Y to be a canonical curve in $\mathbb{P}^{d+1}$, it is necessary and sufficient that

$$\begin{aligned} Y.D &= 2g - 2, \\ Y.(Y + K) &= 2g - 2. \end{aligned}$$

If $n \geq 2e - 2$, then it can be easily checked that $Y = 3C_0 + (n + e + 2)f$ is very ample and satsifies above. So for any $g \geq 4$, the linear system $|3C_0 + (n + e + 2)f|$ with $n = 2g - 2$ and $e = g$ contains a nonsingular curve Y of genus g which is a canonical curve in $\mathbb{P}^{d+1}$. For any such Y, the restriction of the projection map $Y \hookrightarrow X \to \mathbb{P}^1$ is a g^1_3 of Y. Hence, for every $g \geq 4$, there exists a nonhyperelliptic curve which has a g^1_3. □

11. Let X be a ruled surface over the curve C, defined by a normalized bundle $\mathcal{E}$, and let $\mathfrak{e}$ be the divisor on C for which $\mathcal{O}_C(\mathfrak{e}) \cong \bigwedge^2 \mathcal{E}$ (2.8.1). Let $\mathfrak{b}$ be any divisor on C.

(a) If $|\mathfrak{b}|$ and $|\mathfrak{b} + \mathfrak{e}|$ have no base points, and if $\mathfrak{b}$ is nonspecial, then there is a section $D \sim C_0 + \mathfrak{b}f$, and $|D|$ has no base points.

(b) If $\mathfrak{b}$ and $\mathfrak{b} + \mathfrak{e}$ are very ample on C, and for every point $P \in C$, we have $\mathfrak{b} - P$ and $\mathfrak{b} + \mathfrak{e} - P$ nonspecial, then $C_0 + \mathfrak{b}f$ is very ample.

Proof.

(a) Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(\mathfrak{e}) \longrightarrow 0 \ .$$

By the assumption that $\mathfrak{b}$ is nonspecial, we have

$$\mathrm{Ext}^1(\mathcal{O}_C(\mathfrak{e}), \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})) \cong \mathrm{H}^1(\mathcal{O}_C(\mathfrak{b})) = 0,$$

which implies we have a short exact sequence

$$0 \longrightarrow \mathrm{Hom}(\mathcal{O}_C(\mathfrak{e}), \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})) \longrightarrow \mathrm{Hom}(\mathcal{E}, \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})) \longrightarrow \mathrm{Hom}(\mathcal{O}_C, \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})) \longrightarrow 0 \ .$$

Thus, giving a morphism $\mathcal{E} \to \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})$ is equivalent to giving nonzero sections $s \in \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b}))$ and $t \in \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b} + \mathfrak{e}))$, and the corresponding morphism $\mathcal{E} \to \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})$ is locally defined by

$$(f, g) \mapsto fs + gt.$$

Similar to the proof of (2.17), for the corresponding map $\mathcal{E} \to \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})$ to be surjective, it is enough that the divisors corresponding to s and t do not meet. Consider the subset $B \subset X \times |\mathfrak{b}| \times |\mathfrak{b} + \mathfrak{e}|$ consisting of all triples $\langle P, D, E \rangle$ such that P is in the support of both D and E. If $|\mathfrak{b}|$ and $|\mathfrak{b} + \mathfrak{e}|$ are base point

free, then the fibers of $p_1 : B \to X$ have codimension 2 in $|\mathfrak{b}| \times |\mathfrak{b}+\mathfrak{e}|$ by (IV, 3.1), so B has dimension one less than that of $|\mathfrak{b}| \times |\mathfrak{b}+\mathfrak{e}|$. It follows that the image of B under the second projection map $p_2 : B \to |\mathfrak{b}| \times |\mathfrak{b}+\mathfrak{e}|$ is proper, so there exists sections $s \in \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b}))$ and $t \in \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b}+\mathfrak{e}))$ that do not both vanish at any point in C. Hence, we have a short exact sequence of locally free sheaves on C

$$0 \longrightarrow \mathcal{O}_C(-\mathfrak{b}) \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(\mathfrak{b}+\mathfrak{e}) \longrightarrow 0 \ ,$$

so by (2.9), there is a section $D \sim C_0 + \mathfrak{b}f$, and it remains to show $|D|$ has no base points. We will show for every $P \in X$, there exists $D' \in |D|$ such that $P \notin \operatorname{Supp} D'$. Let $\pi : X \to C$ be the projection map, and let $\sigma : C \to X$ be the map corresponding to the section C_0.

$\boxed{P \notin C_0}$ A divisor of the form $D' = C_0 + \mathfrak{b}'f$ where $\mathfrak{b}' \in |\mathfrak{b}|$ misses $\pi(P)$ will do.

$\boxed{P \in C_0}$ Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-C_0) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{C_0} \longrightarrow 0 \ .$$

Tensoring by $\mathcal{O}_X(D)$ and taking cohomology gives

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \xrightarrow{\alpha} \mathrm{H}^0(C_0, \mathcal{O}_{C_0}(\mathfrak{b}+\mathfrak{e})) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X(\mathfrak{b}f)) \ .$$

Since $\mathfrak{b}$ is nonspecial, by (2.4) we have

$$\mathrm{H}^1(X, \mathcal{O}_X(\mathfrak{b}f)) \cong \mathrm{H}^1(C, \mathcal{O}_C(\mathfrak{b})) = 0,$$

so α is a surjection by exactness. This means for any divisor $\mathfrak{b}' \in |\mathfrak{b}+\mathfrak{e}|$ on $C_0 \cong C$, we can find $D' \in |D|$ such that $\sigma^* D' = \mathfrak{b}'$. If $|\mathfrak{b}+\mathfrak{e}|$ has no base points, then we can find a divisor in $|\mathfrak{b}+\mathfrak{e}|$ that misses $P \in C_0$, and we can lift it to a divisor in $|D|$ which also misses P by construction.

Hence, $|D|$ has no base points.

(b) We will show D is very ample by showing that the linear system $|D|$ separates points $P \neq Q$ and tangent vectors $t \in T_P X$. Note that if $\mathfrak{d}$ is any divisor on C, and if there exists a point $P \in C$ such that $\mathfrak{d} - P$ is nonspecial, then $\mathfrak{d}$ is a nonspecial divisor on C. In particular, $\mathfrak{b}$ and $\mathfrak{b}+\mathfrak{e}$ are nonspecial, so we can resume from (a).

$\boxed{\text{Case 1: } P, Q \text{ not both in } C_0, \text{ and not both in any fiber.}}$ Since $\mathfrak{b}$ is very ample, there exists a divisor $\mathfrak{b}' \in |\mathfrak{b}|$ which separates $\pi(P), \pi(Q)$ in C, so $C_0 + \mathfrak{b}'f \in |D|$ separates P, Q in X.

$\boxed{\text{Case 2: } P, t \text{ not both in } C_0, \text{ and not both in any fiber.}}$ Again, $|\mathfrak{b}|$ is very ample, so if t is not in any fiber, then its pushforward $\pi_* t \in T_{\pi(P)} C$ is nonzero. Thus, we can find a divisor of the form $C_0 + \mathfrak{b}'f$ such that $\mathfrak{b}' \in |\mathfrak{b}|$ contains $\pi(P)$ in its support and separates $\pi_* t$ so that $C_0 + \mathfrak{b}'f$ contains P and separates t in X.

$\boxed{\text{Case 3: } P, Q \text{ or } P, t \text{ are both in } C_0.}$ Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-C_0) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{C_0} \longrightarrow 0 \ .$$

Tensoring by $\mathcal{O}_X(D)$ and taking cohomology gives

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \xrightarrow{\beta} \mathrm{H}^0(C_0, \mathcal{O}_X(D) \otimes \mathcal{O}_{C_0}) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X(\mathfrak{b}f)) \ .$$

Since $\mathcal{O}_X(D) \otimes \mathcal{O}_{C_0} \cong \mathcal{O}_{C_0}(\mathfrak{b}+\mathfrak{e})$ is very ample on $C_0 \cong C$, it suffices to show β is a surjection. By (2.4) and the assumption that $|\mathfrak{b}|$ is nonspecial, we have

$$\mathrm{H}^1(X, \mathcal{O}_X(\mathfrak{b}f)) \cong \mathrm{H}^1(C, \pi_* \mathcal{O}_X(\mathfrak{b}f)) \cong \mathrm{H}^1(C, \mathcal{O}_C(\mathfrak{b})) = 0.$$

Hence, β is a surjection by exactness.

$\boxed{\text{Case 4: } P, Q \text{ or } P, t \text{ are both in the same fiber.}}$ Let f be the fiber containing P, Q or P, t. We have the identification $f \sim \pi^*(\pi(P)) = \pi(P)f$. Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-f) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_f \longrightarrow 0 \ .$$

Tensoring by $\mathcal{O}_X(D)$ and taking cohomology gives

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \xrightarrow{\gamma} \mathrm{H}^0(f, \mathcal{O}_X(D) \otimes \mathcal{O}_f) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X(D-f)) \ ,$$

Since $D.f = 1$, the invertible sheaf $\mathcal{O}_X(D) \otimes \mathcal{O}_f$ is very ample on $f \cong \mathbb{P}^1$. Thus, to separate P, Q or P, t, it will be sufficient to show that γ is a surjection. The cokernel of this map lands in $\mathrm{H}^1(X, \mathcal{O}_X(D-f))$, which by (2.4) is isomorphic to

$$\mathrm{H}^1(C, \pi_* \mathcal{O}_X(D-f)).$$

On the other hand, we have

$$\pi_*(\mathcal{O}_X(D-f)) \cong \pi_*(\mathcal{O}_X(C_0)) \otimes \mathcal{O}_C(\mathfrak{b} - \pi(P)) \cong \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b} - \pi(P)).$$

Consider the following exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(\mathfrak{e}) \longrightarrow 0.$$

Tensoring by $\mathcal{O}_C(\mathfrak{b} - \pi(P))$ gives

$$0 \longrightarrow \mathcal{O}_C(\mathfrak{b} - \pi(P)) \longrightarrow \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b} - \pi(P)) \longrightarrow \mathcal{O}_C(\mathfrak{b} + \mathfrak{e} - \pi(P)) \longrightarrow 0 \ ,$$

and taking cohomology gives

$$\mathrm{H}^1(C, \mathcal{O}_C(\mathfrak{b} - \pi(P))) \longrightarrow \mathrm{H}^1(C, \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b} - \pi(P))) \longrightarrow \mathrm{H}^1(C, \mathcal{O}_C(\mathfrak{b} + \mathfrak{e} - \pi(P))).$$

The terms on the left and right are zero since we assumed $\mathfrak{b} - Q$ and $\mathfrak{b} + \mathfrak{e} - Q$ are nonspecial for any $Q \in C$, so the term in the middle is zero by exactness. We conclude that γ is a surjection.
Hence, D is very ample.

□

12. Let X be a ruled surface with invariant e over an elliptic curve C, and let $\mathfrak{b}$ be a divisor on C.

(a) If $\deg \mathfrak{b} \geq e + 2$, then there is a section $D \sim C_0 + \mathfrak{b} f$ such that $|D|$ has no base points.

(b) The linear system $|C_0 + \mathfrak{b} f|$ is very ample if and only if $\deg \mathfrak{b} \geq e + 3$.

Proof. $\boxed{e \geq 0}$ If $\mathfrak{d}$ is a divisor of degree d on C, then $d \geq 1$ if and only if $\mathfrak{d}$ is ample and nonspecial, $d \geq 2$ if and only if $\mathfrak{d}$ is base point free, and $d \geq 3$ if and only if $\mathfrak{d}$ is very ample. If $\deg \mathfrak{b} \geq e + 2 \geq 2$, then $|\mathfrak{b}|$ and $|\mathfrak{b} + \mathfrak{e}|$ have no base points, and $\mathfrak{b}$ is nonspecial, so by (Ex. 2.11) there is a section $D \sim C_0 + \mathfrak{b} f$ such that $|D|$ has no base points.

If $C_0 + \mathfrak{b} f$ is very ample and $e \geq -1$, then $\mathcal{O}_X(C_0 + \mathfrak{b} f) \otimes \mathcal{O}_{C_0}$ is very ample on C_0, which is so if and only if

$$(\deg \mathfrak{b}) - e = C_0.(C_0 + \mathfrak{b} f) = \deg \mathcal{O}_X(C_0 + \mathfrak{b} f) \otimes \mathcal{O}_{C_0} \geq 3.$$

Conversely, if $\deg \mathfrak{b} \geq e + 3$, then $\mathfrak{b}$ and $\mathfrak{b} + \mathfrak{e}$ are very ample on C, and $\mathfrak{b} - P, \mathfrak{b} + \mathfrak{e} - P$ have positive degrees, so they are nonspecial. Hence, by (Ex. 2.11) $C_0 + \mathfrak{b} f$ is very ample.

$\boxed{e = -1}$ Pick a base point $P \in C$ such that $\mathfrak{e} = P$, $X = \mathbb{P}(\mathcal{E})$, and $\mathcal{E}$ fits into the short exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0 \ .$$

For the first statement, if $\deg \mathfrak{b} \geq 2$, then we return to the situation $e \geq 0$, so assume $\mathfrak{b} = Q$ for some $Q \in C$. We want to show there is a section $D \sim C_0 + Qf$ such that $|D|$ has no base points. Take a point $R \in C$ distinct from P and Q such that $2R \sim P + Q$. This is possible, because the linear system $|P + Q|$ defines a two-to-one map of C to $\mathbb{P}^1$, ramified at four points, and we can take R to be one of them. Imitating the proof of (Ex. 2.11), we can find sections in $\mathrm{H}^0(C, \mathcal{O}_C(Q))$ and $\mathrm{H}^0(C, \mathcal{O}_C(P + Q)) \cong \mathrm{H}^0(C, \mathcal{O}_C(2R))$ that do not both vanish at every point in C since $Q \neq R$. Hence, we have a short sequence of locally free sheaves

$$0 \longrightarrow \mathcal{O}_C(-Q) \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P + Q) \longrightarrow 0 \ ,$$

so there is a section $D \sim C_0 + Qf$, and it remains to show $|D|$ has no base points. From now, if x is any point in X, then denote $P_x = \pi(x) \in C$, and if Q is any point in C, then denote $f_Q = \pi^{-1}(Q)$. In particular, $x \in X$ lies on the fiber $f_{\pi(x)}$. We only need to check the points in the support of $C_0 + Qf$, and since $D.f_Q = 1$ and $D.C_0 = 2$, $\mathcal{O}_X(D)$ restricts to a base-point free line bundle on f_Q and C_0, so it is enough to show the natural maps

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \to \mathrm{H}^0(f_Q, \mathcal{O}_X(D) \otimes \mathcal{O}_{f_Q}), \tag{1}$$

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \to \mathrm{H}^0(C_0, \mathcal{O}_X(D) \otimes \mathcal{O}_{C_0}), \tag{2}$$

are surjective. We begin with f_Q. The cokernel of (1) lands in $\mathrm{H}^1(X, \mathcal{O}_X(D - f_Q))$, which by (2.4) is isomorphic to $\mathrm{H}^1(C, \pi_*\mathcal{O}_X(D - f_Q))$. On the other hand, $D - f_Q \sim C_0$, so

$$\pi_*\mathcal{O}_X(D - f_Q) \cong \mathcal{E},$$

and $\mathrm{H}^1(C, \mathcal{E}) = 0$ since $\mathcal{E}$ is the unique nontrivial extension of $\mathcal{O}_C(P)$ by $\mathcal{O}_C$. Therefore, (1) is surjective. Moving on to C_0, the cokernel of (2) lands in $\mathrm{H}^1(X, \mathcal{O}_X(D - C_0))$, which by (2.4) is isomorphic to $\mathrm{H}^1(C, \pi_*\mathcal{O}_X(D - C_0))$. On the other hand, $D - C_0 \sim f_Q$, so

$$\pi_*\mathcal{O}_X(D - C_0) \cong \mathcal{O}_C(Q),$$

which is nonspecial. Therefore, $\mathrm{H}^1(X, \mathcal{O}_X(D - C_0)) = 0$, and (2) is surjective.

Now suppose $\mathfrak{b}$ is a divisor on C with $\deg \mathfrak{b} \geq e + 3 = 2$. If $\deg \mathfrak{b} \geq 3$, then we can apply the same argument as in $e \geq 0$, so assume $\deg \mathfrak{b} = 2$, which means $\mathfrak{b}$ is a g^1_2 on C. Let $x \neq y$ be points in X, and let t be a tangent vector at x. We will show $D = C_0 + \mathfrak{b}f$ is very ample by showing that the linear system $|D|$ separates points and tangent vectors.

Case 1: x, y not both in C_0, and not both in any fiber. Without loss of generality, assume $y \notin C_0$, and let Q_x be a point in C such that $P_x + Q_x \in |\mathfrak{b}|$. If $P_y \neq Q_x$, then the divisor $C_0 + (P_x + Q_x)f$ separates x and y. Otherwise, we have $P_x + P_y \in |\mathfrak{b}|$, and for any point $z \in X$, consider the section $C_z = C_0 + (P_z - P)f$ (Ex. 2.7). Note that if $P_z \neq P$, then $C_0 \cap C_z = P_z$. If $P = P_x$, then $C_y + 2f_P \in |D|$ goes through x but not y. The hard case is $P = P_y$, and we further divide to two cases.

Case 1a. $x \in C_0$ Since $y \notin C_0$, let $S + T \in |\mathfrak{b}|$ such that $S, T \neq P_y$, then then divisor $C_0 + (S+T)f \in |D|$ goes through x but not y.

Case 1b. $x \notin C_0$ The section $C_x = C_0 + (P_x - P)f$ does not pass through x, and since $2P = 2P_y \in |\mathfrak{b} + P - P_x|$, the section $C_x + 2f_P \in |D|$ goes through y but not x.

Case 2: x, t not both in C_0, and not both in any fiber. A divisor of the form $C_0 + (P_x + Q_x)f$ will do.

Case 3: x, y or x, t are both in C_0. Since $C_0.D = 3$, the invertible sheaf $\mathcal{O}_X(D) \otimes \mathcal{O}_{C_0}$ is very ample on $C_0 \cong C$, so it is enough to show the natural map

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \to \mathrm{H}^0(C_0, \mathcal{O}_X(D) \otimes \mathcal{O}_{C_0})$$

is surjective. The cokernel of this map lands in $\mathrm{H}^1(X, \mathcal{O}_X(D - C_0))$, which by (2.4) is isomorphic to $\mathrm{H}^1(C, \pi_*\mathcal{O}_X(D - C_0))$. On the other hand, $D - C_0 \sim \mathfrak{b}f$, so

$$\pi_*\mathcal{O}_X(D - C_0) \cong \mathcal{O}_C(\mathfrak{b}),$$

which is nonspecial on C. Hence, $\mathrm{H}^1(X, \mathcal{O}_X(D - C_0)) = 0$, and the map above is surjective.

Case 4: x, y or x, t are both in the same fiber $f = f_Q$ for some $Q \in C$. Since $D.f = 1$, the invertible sheaf $\mathcal{O}_X(D) \otimes \mathcal{O}_f$ is very ample on $f \cong \mathbb{P}^1$, so it is enough to show the natural map

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \to \mathrm{H}^0(f, \mathcal{O}_X(D) \otimes \mathcal{O}_f)$$

is surjective. The cokernel of this map lands in $\mathrm{H}^1(X, \mathcal{O}_X(D - f))$, which by (2.4) is isomorphic to $\mathrm{H}^1(C, \pi_*\mathcal{O}_X(D - f))$. On the other hand, $D - f \sim C_0 + (\mathfrak{b} - Q)f$, so if we write $R \sim \mathfrak{b} - Q$ for some $R \in C$, then

$$\pi_*\mathcal{O}_X(D - f) \cong \mathcal{E} \otimes \mathcal{O}_C(R)$$

by the projection formula. We have a short exact sequence

$$\mathrm{H}^1(C, \mathcal{O}_C(R)) \longrightarrow \mathrm{H}^1(C, \mathcal{E} \otimes \mathcal{O}_C(R)) \longrightarrow \mathrm{H}^1(C, \mathcal{O}_C(P+R)) \ ,$$

and since the terms on the left and right are zero, the term in the middle is zero. Hence, $\mathrm{H}^1(X, \mathcal{O}_X(D - f)) = 0$, and the map above is surjective. □

13. For every $e \geq -1$ and $n \geq e+3$, there is an elliptic scroll of degree $d = 2n - e$ in $\mathbb{P}^{d-1}$. In particular, there is an elliptic scroll of degree 5 in $\mathbb{P}^4$.

Proof. For any divisor $\mathfrak{b}$ on C of degree n, we use the divisor $D = C_0 + \mathfrak{b}f$. By (Ex. 2.12), $C_0 + \mathfrak{b}f$ is very ample, and since $D.f = 1$ for any fiber f, the image of X in $|D| \cong \mathbb{P}^N$ is a scroll. The degree of the image is

$$D^2 = 2(\deg \mathfrak{b}) - e = 2n - e,$$

and it remains to compute N. By (2.4), we find that

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) = \mathrm{H}^0(C, \pi_* \mathcal{O}_X(D)) = \mathrm{H}^0(C, \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b})),$$

and as in the proof of (Ex. 2.12), we have a short exact sequence

$$0 \longrightarrow \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b})) \longrightarrow \mathrm{H}^0(C, \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b})) \longrightarrow \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b}+P)) \longrightarrow 0.$$

Hence, we can compute N using exactness of above

$$\begin{aligned} N &= h^0(C, \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b})) - 1 \\ &= h^0(C, \mathcal{O}_C(\mathfrak{b})) + h^0(C, \mathcal{O}_C(\mathfrak{b}+\mathfrak{e})) - 1 \\ &= 2(\deg \mathfrak{b}) + \deg \mathfrak{e} - 1 \\ &= 2n - e - 1. \end{aligned}$$

To obtain an elliptic scroll of degree 5 in $\mathbb{P}^4$, take $n = 2$ and $e = -1$. □

14. Let X be a ruled surface over a curve of genus g, with invariant $e < 0$, and assume that $\operatorname{char} k = p > 0$ and $g \geq 2$.

(a) If $Y \equiv aC_0 + bf$ is an irreducible curve $\neq C_0, f$, then either $a = 1$, $b \geq 0$, or $2 \leq a \leq p-1$, $b \geq \frac{1}{2}ae$, or $a \geq p$, $b \geq \frac{1}{2}ae + 1 - g$.

(b) If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g-1))$, then any divisor $D \equiv aC_0 + bf$ is ample. On the other hand, if D is ample, then $a > 0$ and $b > \frac{1}{2}ae$.

Proof.

(a) Let $\tilde{Y}$ be the normalization of Y, and consider the composition of the natural map $\tilde{Y} \to Y$ with the projection $\pi : Y \to C$. If $a \leq p-1$, then π is separable, so we can apply the same argument as in (2.21). If $a \geq p$, then $\pi : \tilde{Y} \to C$ may not be separable. By (IV, 2.5.4), $\pi : \tilde{Y} \to C$ factors as the composite $\tilde{Y} \to Z \to C$, where Z is a smooth curve over k, $\tilde{Y} \to Z$ is purely inseparable of degree p^n for some $n \geq 0$, and $Z \to C$ is finite, separable of degree d such that $a = dp^n$. By Hurwitz's theorem and the fact that $p_a(Y) \geq g(\tilde{Y})$, we have

$$2p_a(Y) - 2 \geq 2g(\tilde{Y}) - 2 = 2g(Z) - 2 \geq d(2g-2).$$

and the adjunction formula (1.5) gives

$$2p_a(Y) - 2 = Y.(Y + K),$$

where K is a canonical divisor of X. Substituting $Y \equiv aC_0 + bf$ and $K \equiv -2C_0 + (2g-2-e)f$ from (2.11), and combining with the inequality above, we find that

$$b(a-1) \geq \frac{1}{2}ae(a-1) + (d-a)(g-1).$$

If $a \geq p$, then $\frac{d-a}{a-1} \geq -1$, so

$$b \geq \frac{1}{2}ae + 1 - g.$$

(b) If D is ample, then by (1.10) $D.f = a > 0$, and $D^2 = 2ab - a^2e > 0$, so $b > \frac{1}{2}ae$. If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g-1))$, then $D.f = a > 0$, $D.C_0 = a((1/p)(g-1) - \frac{1}{2}e) > 0$, and

$$D^2 = a(2b - ae) > \frac{2a}{p}(g-1) > 0.$$

If $Y \equiv a'C_0 + b'f$ is any irreducible curve $\neq C_0, f$, then

$$D.Y = ab' + a'b - aa'e.$$

Note that $b > \frac{1}{2}ae$, so the same argument as in (2.21) for when $1 \leq a' \leq p-1$ applies here as well. Now if $a' = 1$, then $b' \geq 0$, so

$$D.Y > \frac{1}{2}ae - ae = -\frac{1}{2}ae > 0.$$

If $2 \leq a' \leq p-1$, then $b' \geq \frac{1}{2}a'e$, so

$$D.Y > \frac{1}{2}aa'e + \frac{1}{2}aa'e - aa'e = 0.$$

If $a' \geq p$, then $b' \geq \frac{1}{2}a'e + 1 - g$, so

$$D.Y > a(\frac{1}{2}a'e + 1 - g) + a'a(\frac{1}{2}e + (1/p)(g-1)) - aa'e = \left(\frac{a'}{p} - 1\right)a(g-1) > 0.$$

Therefore by (1.10), D is ample.

□

16. Let C be a nonsingular affine curve. Show that two locally free sheaves $\mathcal{E}, \mathcal{E}'$ of the same rank are isomorphic if and only if their classes in the Grothendieck group $K(X)$ are the same. This is false for a projective curve.

Proof. Let A be a be an affine coordinate ring of a nonsingular affine curve, and let K be its function field. The category of coherent $\mathcal{O}_C$-modules is equivalent to the category of finitely generated A-modules. Under this identification, $K(X)$ is naturally isomorphic to the Grothendieck group of flat finitely generated A-modules $K_0(A)$ (A.M., Ex. 7.26). The structure theorem of modules over Dedekind domains states $K_0(A) \cong \mathbb{Z} \oplus \mathrm{Cl}(A)$, where if M is any flat finitely generated A-module, then $M \cong A^r \oplus I$, with $r + 1 = \dim_K M \otimes_A K$, and I is some fractional ideal in $\mathrm{Cl}(A)$. If $N \cong A^s \oplus J$ is another A-module, then $M \cong N$ if and only if $r = s$ and $I \cong J$ in $\mathrm{Cl}(A)$. □

17. (a) Let $\varphi : \mathbb{P}^1_k \to \mathbb{P}^3_k$ be the 3-uple embedding. Let $\mathcal{I}$ be the sheaf of ideals of the twisted cubic curve C which is the image of φ. Then $\mathcal{I}/\mathcal{I}^2$ is a locally free sheaf of rank 2 on C, so $\varphi^*(\mathcal{I}/\mathcal{I}^2)$ is a locally free sheaf of rank 2 on $\mathbb{P}^1$. By (2.14), therefore, $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$ for some $l, m \in \mathbb{Z}$. Determine l and m.

(b) Repeat part (a) for the embedding $\varphi : \mathbb{P}^1 \to \mathbb{P}^3$ given by $x_0 = t^4, x_1 = t^3u, x_2 = tu^3, x_3 = u^4$, whose image is a nonsingular rational quartic curve.

Proof.

(a) Let t, u be homogenous coordinates for C, and let x_0, x_1, x_2, x_3 be homogenous coordinates for $\mathbb{P}^3$, so that φ is expicitly given by

$$(t : u) \mapsto (t^3 : t^2u : tu^2 : u^3).$$

It is enough to determine $\mathcal{N}_{C/\mathbb{P}^3} = \varphi^*(\mathcal{I}/\mathcal{I}^2)^\vee$. If Q is any irreducible quadric surface containing C in $\mathbb{P}^3$, then C has type $(1,2)$ as a divisor on Q, and we can use the short exact sequence of normal bundles

$$0 \longrightarrow \mathcal{N}_{C/Q} \longrightarrow \mathcal{N}_{C/\mathbb{P}^3} \longrightarrow \mathcal{N}_{Q/\mathbb{P}^3}\big|_C \longrightarrow 0,$$

where we have

$$\mathcal{N}_{C/Q} \cong \mathcal{O}_C(4), \quad \mathcal{N}_{Q/\mathbb{P}^3}\big|_C \cong \mathcal{O}_C(6).$$

By exactness, we have two possibilities for the isomorphism class of $\mathcal{N}_{C/\mathbb{P}^3}$, which are distinguished by whether $\mathcal{N}_{C/\mathbb{P}^3}(-6)$ has global sections or not. In particular, we have

$$\mathcal{N}_{C/\mathbb{P}^3} \cong \begin{cases} \mathcal{O}_C(5) \oplus \mathcal{O}_C(5), & \text{if } h^0(\mathcal{N}_{C/\mathbb{P}^3}(-6)) = 0, \\ \mathcal{O}_C(4) \oplus \mathcal{O}_C(6), & \text{otherwise.} \end{cases}$$

The two relevant short exact sequences for this computation are

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_{\mathbb{P}^3}(1)^{\oplus 4} \longrightarrow \mathcal{T}_{\mathbb{P}^3} \longrightarrow 0 \quad (*)$$

$$0 \longrightarrow \mathcal{T}_C \longrightarrow \mathcal{T}_{\mathbb{P}^3}\big|_C \longrightarrow \mathcal{N}_{C/\mathbb{P}^3} \longrightarrow 0 \quad (**)$$

Restricting to C, twisting by $\mathcal{O}_C(-6)$, and taking the long exact sequence in cohomology of (*) gives

$$0 \longrightarrow \mathrm{H}^0(\mathcal{O}_C(-6)) \longrightarrow \mathrm{H}^0(\mathcal{O}_C(-3)^{\oplus 4}) \longrightarrow \mathrm{H}^0(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) \longrightarrow$$

$$\longrightarrow \mathrm{H}^1(\mathcal{O}_C(-6)) \xrightarrow{\beta} \mathrm{H}^1(\mathcal{O}_C(-3)^{\oplus 4}) \longrightarrow \mathrm{H}^1(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) \longrightarrow 0.$$

The first two terms on the top row vanish since $\mathrm{H}^0(\mathcal{O}_C(-n)) = 0$ for any $n > 0$, the first two terms on the bottom row can be computed using Serre duality

$$h^1(\mathcal{O}_C(-6)) = h^0(\mathcal{O}_C(4)) = 5, \quad h^1(\mathcal{O}_C(-3)^{\oplus 4}) = h^0(\mathcal{O}_C(1)^{\oplus 4}) = 8,$$

and β is dual to

$$\beta' : \mathrm{H}^0(\mathcal{O}_C(1)^{\oplus 4}) \to \mathrm{H}^0(\mathcal{O}_C(4)),$$

which is explicitly given by

$$(g_1, g_2, g_3, g_4) \mapsto g_1 t^3 + g_2 t^2 u + g_3 t u^2 + g_4 u^3.$$

It is not hard to see β' is surjective, which means β is injective. So by exactness,

$$h^0(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) = 0, \quad h^1(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) = 3.$$

Doing the same calculation for (**) gives

$$0 \longrightarrow \mathrm{H}^0(\mathcal{T}_C(-6)) \longrightarrow \mathrm{H}^0(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) \longrightarrow \mathrm{H}^0(\mathcal{N}_{C/\mathbb{P}^3}(-6)) \longrightarrow$$

$$\longrightarrow \mathrm{H}^1(\mathcal{T}_C(-6)) \xrightarrow{\alpha} \mathrm{H}^1(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) \longrightarrow \mathrm{H}^1(\mathcal{N}_{C/\mathbb{P}^3}(-6)) \longrightarrow 0.$$

The first two terms on the top row vanish, and by Serre duality,

$$h^1(\mathcal{T}_C(-6)) = h^0(\Omega_C(4)) = 3,$$

and α is dual to the map

$$\alpha' : \mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(4)) \to \mathrm{H}^0(\Omega_C(4)).$$

The vector space $\mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(4))$ can be identified with $\ker \beta'$, and $\mathrm{H}^0(\Omega_C(4)) \cong \mathrm{H}^0(\mathcal{O}_C(2))$. This means $\mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(4))$ is generated by the syzygies of β', so we can write a basis

$$s_0 = u\mathrm{d}x_0 - t\mathrm{d}x_1, \quad s_1 = u\mathrm{d}x_1 - t\mathrm{d}x_2, \quad s_2 = u\mathrm{d}x_2 - t\mathrm{d}x_3.$$

Now, we can compute the effect of α' on each s_i,

$$\begin{aligned} \alpha'(s_0) &= u\,\mathrm{d}(t^3) - t\,\mathrm{d}(t^2 u) = t^2(u\mathrm{d}t - t\mathrm{d}u), \\ \alpha'(s_1) &= u\,\mathrm{d}(t^2 u) - t\,\mathrm{d}(t u^2) = tu(u\mathrm{d}t - t\mathrm{d}u), \\ \alpha'(s_2) &= u\,\mathrm{d}(t u^2) - t\,\mathrm{d}(u^3) = u^2(u\mathrm{d}t - t\mathrm{d}u), \end{aligned}$$

so α is an isomorphism. Hence, we have

$$\mathcal{N}_{C/\mathbb{P}^3} \cong \mathcal{O}_C(5) \oplus \mathcal{O}_C(5).$$

(b) We can do the same calculation as above, except in characteristic 2, the corresponding α' has a nontrivial kernel of rank 1. Explicitly, we have

$$\alpha' : \mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(6)) \to \mathrm{H}^0(\Omega_C(6)),$$

where we can identify $\Omega_C(6) \cong \mathcal{O}_C(4)$, and

$$\mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(6)) = \ker(\beta' : \mathrm{H}^0(\mathcal{O}_C(2)^{\oplus 4}) \to \mathrm{H}^0(\mathcal{O}_C(6))).$$

Consider the section of $\Omega_{\mathbb{P}^3}\big|_C(6)$ defined by

$$s = u^2 \mathrm{d}x_1 - t^2 \mathrm{d}x_2.$$

Its image under β can be computed by plugging in $x_2 = t^3 u$ and $x_3 = tu^3$,

$$\begin{aligned}\beta'(s) &= u^2\,\mathrm{d}\big(t^3 u\big) - t^2\,\mathrm{d}\big(tu^3\big) \\ &= u^2(3t^2 u\mathrm{d}t + t^3\mathrm{d}u) - t^2(u^3\mathrm{d}t + 3tu^2\mathrm{d}u) \\ &= 2t^2u^2(u\mathrm{d}t - t\mathrm{d}u),\end{aligned}$$

which is zero if $\operatorname{char} k = 2$. To summarize,

$$\mathcal{N}_{C/\mathbb{P}^3} = \begin{cases}\mathcal{O}_C(6) \oplus \mathcal{O}_C(8) & \operatorname{char} k = 2, \\ \mathcal{O}_C(7) \oplus \mathcal{O}_C(7) & \text{otherwise.}\end{cases}$$

□